Web-Based Application Development

Lab Report CSE 3023 (K1)

Name: Muhammad Syukran Bin Mohd Zakri
Number Matrics: S75572
Lecturer's Name: Prof. Madya Ts. Dr. Wan Nural Jawahir binti Wan Yussof
Ms. Wan Fatin Fatihah binti Yahya
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Exercise Lab 2

ProductDAO.java

```java
package com.lab.dao;

/**
 *
 * @author MSMZ
 */
import com.lab.model.Product;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.util.ArrayList;
import java.util.List;

public class ProductDAO {
    private String jdbcURL = "jdbc:mysql://localhost:3306/CSE3023";
    private String jdbcUsername = "root";
    private String jdbcPassword = "";

    protected Connection getConnection() {
    Connection connection = null;
    try {
        Class.forName("com.mysql.cj.jdbc.Driver");
        connection = DriverManager.getConnection(jdbcURL,jdbcUsername, jdbcPassword);
    } catch (SQLException | ClassNotFoundException e) {
        e.printStackTrace();
    }
    return connection;
}
    // INSERT
    public void insertProduct(Product product) {
    String sql = "INSERT INTO products (name, category, price, quantity) VALUES (?, ?, ?, ?)";
        try (Connection conn = getConnection();
            PreparedStatement pstmt = conn.prepareStatement(sql)) {

            pstmt.setString(1, product.getName());
            pstmt.setString(2, product.getCategory());
            pstmt.setDouble(3, product.getPrice());
            pstmt.setInt(4, product.getQuantity());

            pstmt.executeUpdate();
        } catch (SQLException e) {
            e.printStackTrace();
    }
    //SELECT
    public Product selectProduct(int id) {
    Product product = null;
    String sql = "SELECT name, category, price, quantity FROM products WHERE id =?";
    try (Connection conn = getConnection();
        PreparedStatement pstmt = conn.prepareStatement(sql)) {
       pstmt.setInt(1, id);
       ResultSet rs = pstmt.executeQuery();

       if (rs.next()) {
                String name = rs.getString("name");
                String category = rs.getString("category");
                double price = rs.getDouble("price");
                int quantity = rs.getInt("quantity");
                product = new Product(id, name, category, price, quantity);
       }
    } catch (SQLException e) {
        e.printStackTrace();
    }
    return product;
}
public List<Product> selectAllProduct() {
    List<Product> product = new ArrayList<>();
    String sql = "SELECT * FROM products";
    try (Connection conn = getConnection();
        PreparedStatement pstmt = conn.prepareStatement(sql)) {
        ResultSet rs = pstmt.executeQuery();
        while (rs.next()) {
                int id = rs.getInt("id");
                String name = rs.getString("name");
                String category = rs.getString("category");
                double price = rs.getDouble("price");
                int quantity = rs.getInt("quantity");

                product.add(new Product(id, name, category, price, quantity));
            }
    }catch (SQLException e){
        e.printStackTrace();
    }
    return product;
}
//UPDATE
public boolean updateProduct(Product product) {
```

```java
//UPDATE
public boolean updateProduct(Product product) {
    boolean rowUpdated = false;
    String sql = "UPDATE products SET name = ?, category = ?,price = ?, quantity= ? WHERE id = ?";
    try (Connection conn = getConnection();
        PreparedStatement pstmt = conn.prepareStatement(sql)) {
        pstmt.setString(1, product.getName());
        pstmt.setString(2, product.getCategory());
        pstmt.setDouble(3, product.getPrice());
        pstmt.setInt(4, product.getQuantity());
        pstmt.setInt(5, product.getId());
        rowUpdated = pstmt.executeUpdate() > 0;
    } catch (SQLException e) {
        e.printStackTrace();
    }
    return rowUpdated;
}
// DELETE
    public boolean deleteProduct(int id) {
    boolean rowDeleted = false;
    String sql = "DELETE FROM products WHERE id = ?";
    try (Connection conn = getConnection();
        PreparedStatement pstmt = conn.prepareStatement(sql)) {
        pstmt.setInt(1, id);
        rowDeleted = pstmt.executeUpdate() > 0;
    } catch (SQLException e) {
        e.printStackTrace();
    }
    return rowDeleted;
    }
}
```

Product.java

```java
public Product(int id, String name, String category, double price, int quantity) {
    this.id = id;
    this.name = name;
    this.category = category;
    this.price = price;
    this.quantity = quantity;
}
public Product(String name, String category, double price, int quantity) {
    this.name = name;
    this.category = category;
    this.price = price;
    this.quantity = quantity;
}
// Getter and Setter
public int getId() {
    return id;
}
public void setId(int id) {
    this.id = id;
}
public String getName() {
    return name;
}
public void setName(String name) {
    this.name = name;
}
public String getCategory() {
    return category;
}
public void setCategory(String category) {
    this.category = category;
}
public double getPrice() {
    return price;
}
public void setPrice(double price) {
    this.price = price;
}
public int getQuantity() {
    return quantity;
}
public void setQuantity(int quantity) {
    this.quantity = quantity;
}
```

Add_product.html

```html
<!DOCTYPE html>
<html lang="en">
<!--
Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
Click nbfs://nbhost/SystemFileSystem/Templates/JSP_Servlet/Html.html to edit this template
-->
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Login Page</title>
</head>
<body>
    <h2>Add Product</h2>
    <form action="InsertProductServlet" method="POST">
        <label for="name">Product Name:</label>
        <input type="text" id="name" name="name" required><br><br>
        <label for="category">Category:</label>
        <input type="text" id="category" name="category" required><br><br>
        <label for="price">Price:</label>
        <input type="text" id="price" name="price" required><br><br>
        <label for="quantity">Quantity:</label>
        <input type="text" id="quantity" name="quantity" required><br><br>
        <input type="submit" value="Add Product">
    </form>
    <br>
    <a href="ViewProductServlet">View Products</a>
    <a href='UpdateProductServlet?id=1'>Edit</a> |
    <a href='DeleteProductServlet?id=1'>Delete</a>
</body>
</html>
```

InsertProductServlet

```java
import com.lab.dao.ProductDAO;
import com.lab.model.Product;
import java.io.IOException;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

/**
 *
 * @author MSMZ
 */
public class InsertProductServlet extends HttpServlet {

    private ProductDAO productDAO;

    public void init(){
        productDAO = new ProductDAO();
    }

    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        String name = request.getParameter("name");
        String category = request.getParameter("category");
        double price = Double.parseDouble(request.getParameter("price"));
        int quantity = Integer.parseInt(request.getParameter("quantity"));

        Product newProduct = new Product(name, category, price, quantity);
        productDAO.insertProduct(newProduct); // Panggil DAO, bukan JDBC
        response.sendRedirect("ViewProductServlet");
    }
}
```

ViewProductServlet

```java
import java.io.PrintWriter;
import java.util.List;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
/**
 *
 * @author MSMZ
 */
public class ViewProductServlet extends HttpServlet {
        private ProductDAO productDAO;
        public void init() {
            productDAO = new ProductDAO();
        }
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        List<Product> listProduct = productDAO.selectAllProduct(); // Panggil DAO
        out.println("<h2>Product List (Using DAO)</h2>");
        out.println("<table border='1'><tr><th>ID</th><th>Name</th><th>Category</th><th>Price</th><th>Quantity</th><th>Actions</th></tr>");

        for (Product product : listProduct) {
            out.println("<tr>");
            out.println("<td>" + product.getId() + "</td>");
            out.println("<td>" + product.getName() + "</td>");
            out.println("<td>" + product.getCategory() + "</td>");
            out.println("<td>" + product.getPrice() + "</td>");
            out.println("<td>" + product.getQuantity() + "</td>");
            out.println("<td>");
            out.println("<a href='UpdateProductServlet?id=" + product.getId() + "'>Edit</a> | ");
            out.println("<a href='DeleteProductServlet?id=" + product.getId() + "'>Delete</a>");
            out.println("</td>");
            out.println("</tr>");
        }

        out.println("</table>");
        out.println("<br><a href='index.html'>Add New Product</a>");
    }
}
```

UpdateProductServlet

```java
import com.lab.dao.ProductDAO;
import com.lab.model.Product;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
/**
 *
 * @author MSMZ
 */

public class UpdateProductServlet extends HttpServlet {
    private ProductDAO productDAO;

    public void init(){
        productDAO = new ProductDAO();
    }

    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        int id = Integer.parseInt(request.getParameter("id"));
        Product existingProduct = productDAO.selectProduct(id); // Panggil DAO

        if (existingProduct != null) {
            out.println("<h2>Update Product</h2>");
            out.println("<form action='UpdateProductServlet' method='POST'>");

            out.println("<input type='hidden' name='id' value='" + existingProduct.getId() + "'>");

            out.println("Name: <input type='text' name='name' value='" + existingProduct.getName() + "' required><br><br>");
            out.println("Category: <input type='text' name='category' value='" + existingProduct.getCategory() + "' required><br><br>");
            out.println("Price: <input type='text' name='price' value='" + existingProduct.getPrice() + "' required><br><br>");
            out.println("Quantity: <input type='text' name='quantity' value='" + existingProduct.getQuantity() + "' required><br><br>");

            out.println("<input type='submit' value='Update Product'>");
            out.println("</form>");
        }
    }
```

```java
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        int id = Integer.parseInt(request.getParameter("id"));
        String name = request.getParameter("name");
        String category = request.getParameter("category");
        double price = Double.parseDouble(request.getParameter("price"));
        int quantity = Integer.parseInt(request.getParameter("quantity"));

        Product product = new Product(id, name, category, price, quantity);
        productDAO.updateProduct(product); // Panggil DAO
        response.sendRedirect("ViewProductServlet");
    }
}
```

DeleteProductServlet

```java
package com.lab.controller;

import com.lab.dao.ProductDAO;
import java.io.IOException;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
/**
 *
 * @author MSMZ
 */

public class DeleteProductServlet extends HttpServlet {
    private ProductDAO productDAO;

    public void init(){
        productDAO = new ProductDAO();
    }

    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        int id = Integer.parseInt(request.getParameter("id"));
        productDAO.deleteProduct(id); // Panggil DAO
        response.sendRedirect("ViewProductServlet");
    }
}
```

Output

Product List (Using DAO)

ID	Name	Category	Price	Quantity	Actions
2	Macbook	computer	4000.0	1	Edit | Delete
3	RTX6060	accessories	4000.0	1	Edit | Delete
6	Iphone 15	smartphone	3000.0	3	Edit | Delete
7	Arai	Helmet	2500.0	2	Edit | Delete

Add New Product

Update Product

Name: Macbook

Category: computer

Price: 4000.0

Quantity: 1

Update Product

Delete product

Product List (Using DAO)

ID	Name	Category	Price	Quantity	Actions
2	Macbook	computer	4000.0	1	Edit | Delete
6	Iphone 15	smartphone	3000.0	3	Edit | Delete
7	Arai	Helmet	2500.0	2	Edit | Delete
8	Macbook	computer	4000.0	1	Edit | Delete

Add New Product